STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-SCHEMES-MORPHISMS-RS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1741-1747 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 1749-1775, the complete author proof (25 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 32 of queue_examples-defos.json (pb=948, stmt 1741-1747, proof 1749-1775).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0E3T: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1714–1731. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06J5: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1775–1775. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07RX: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1775–1775. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07RT: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1775–1775. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Example 0E3T satisfies the Rim-Schlessinger condition (RS). In particular, $\mathcal{D}ef_{X\to Y}$ is a deformation category for any morphism of schemes $X \to Y$ over k.*

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $A_1 \to A$ and $A_2 \to A$ be morphisms of $\mathcal{C}_\Lambda$. Assume $A_2 \to A$ is surjective. According to Formal Deformation Theory, Lemma 06J5 it suffices to show that the functor $\mathcal{F}(A_1 \times_A A_2) \to \mathcal{F}(A_1) \times_{\mathcal{F}(A)} \mathcal{F}(A_2)$ is an equivalence of categories. Observe that

$$\begin{array}{ccc} \mathrm{Spec}(A) & \longrightarrow & \mathrm{Spec}(A_2) \\ \downarrow & & \downarrow \\ \mathrm{Spec}(A_1) & \longrightarrow & \mathrm{Spec}(A_1 \times_A A_2) \end{array}$$

is a pushout diagram as in More on Morphisms, Lemma 07RT. Thus the lemma follows immediately from More on Morphisms, Lemma 07RX as this describes the category of schemes flat over $A_1 \times_A A_2$ as the fibre product of the category of schemes flat over A_1 with the category of schemes flat over A_2 over the category of schemes flat over A. $\square$